



J. Dairy Sci. TBC

<https://doi.org/10.3168/jds.2024-25979>

© TBC, The Authors. Published by Elsevier Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (<https://creativecommons.org/licenses/by/4.0/>).

Associations between periparturient calcium dynamics of multiparous Holstein cows and inflammation markers during the transition period

G. M. Graef,¹ A. Sipka,² S. Tompkins,¹ C. R. Seely,³ J. A. A. McArt,² and T. R. Overton^{1*}

¹Department of Animal Science, Cornell University, Ithaca, NY 14850

²Population Medicine and Diagnostic Sciences, Cornell University, Ithaca, NY 14850

³Department of Agriculture, Nutrition, and Food Systems, University of New Hampshire, Durham, NH 03824

ABSTRACT

Recent work has established that proper classification of cows based on their periparturient status has important implications for risks of health, production, and reproduction outcomes during the postpartum period. However, the dynamics of blood calcium (Ca) status and inflammation are far less understood. The objective of this study was to determine the relationship between subclinical hypocalcemia (SCH) dynamics and periparturient circulating serum amyloid-A (SAA), haptoglobin (Hp), tumor necrosis factor- α (TNF α), interferon- γ (IFN γ), and interleukin-10 (IL-10) concentrations, DMI, and milk production during the first 9 wk of lactation. Data collected from multiparous Holstein cows ($n = 96$) were retrospectively classified into 1 of 4 calcemia groups based on blood concentrations of total Ca (tCa) at 1 and 5 DIM: normocalcemic (NC; [tCa] >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH; [tCa] ≤ 1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH; [tCa] >1.95 mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH; [tCa] ≤ 1.95 mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$). The MIXED procedure of SAS (v. 9.4, SAS Institute Inc.) was used to analyze pre- (-28 and -1 d relative to calving) and postpartum (1, 3, 5, and 7 DIM) blood mineral concentrations, SAA, Hp, cytokines, milk production, and differences between Ca groups, using repeated measures for time. Prepartum blood concentrations of IL-10 tended to be greater in tSCH cows compared with NC, dSCH, and pSCH. Postpartum concentrations of Hp, SAA, and TNF α were different by Ca group. Concentrations of SAA and Hp were highest in the pSCH group at d 3, and TNF α was highest for cows categorized as dSCH, irrespective of time. Overall, our results suggest that cows experienc-

ing different SCH dynamics display different patterns of inflammatory markers and that elevated concentrations of proinflammatory biomarkers accompany some categories of hypocalcemia during the early lactation period.

Key words: subclinical, dyscalcemia, transition, inflammation

INTRODUCTION

The transition from late gestation to the onset of early lactation is a critical point of adaptation to rapid metabolic changes. This period of transition for the dairy cow presents a set of challenges, and her success is dependent upon her ability to adapt efficiently. This time is typically marked by a significant decline in DMI prepartum and postpartum DMI, failing to meet the energetic and nutrient demands of milk production (Bauman and Currie, 1980). Simultaneously, rapid increases in Ca demands are often cited as particularly stringent with a 2- to 3-fold surge in demand at parturition (Drackley et al., 2005; Goff et al., 2014). Clinical hypocalcemia is relatively well-controlled in confinement dairy systems (Reinhardt et al., 2011). The subclinical iteration of hypocalcemia, subclinical hypocalcemia (SCH), is more prevalent, and its presence past 1 DIM is considered to put cows at increased risk for other periparturient disorders, decreased postpartum DMI, poor reproductive performance, and milk production losses (Neves et al., 2018; McArt and Neves, 2020; Seely and McArt, 2022). With this in mind, recent work has established 4 distinct categories of SCH in cows based upon their lactation, health and culling outcomes, and postpartum Ca dynamics on 1, 2, and 4 DIM (dependent on parity; McArt and Neves, 2020). These groups, normocalcemia (NC), transient SCH (tSCH), delayed SCH (dSCH), and persistent SCH (pSCH), are notably different in their relative risks for disease, production, and reproductive outcomes (Neves et al., 2018; McArt and Neves, 2020).

Received November 7, 2024.

Accepted July 4, 2025.

*Corresponding author: tro2@cornell.edu

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-25. Nonstandard abbreviations are available in the Notes.

As gestation comes to an end and lactogenesis begins, acute inflammation is considered normal if it is limited in duration and severity (Bradford and Swartz, 2020). Released from macrophages and tissues, inflammatory mediators are intimately involved in the rapid metabolic changes occurring during transition, including increased lipid mobilization, alterations in endocrine function, and increased oxidative stress (Sordillo and Aitken, 2009; Trevisi et al., 2015). However, an uncontrolled or unresolved inflammatory response can be detrimental to a successful transition for the cow, resulting in increased risk for periparturient disease, increased culling, and decreased production (Drackley, 1999; Bertoni et al., 2015; Bradford et al., 2015). Whereas SCH is understood to increase risk of other disease events, it has also been observed that rapid declines in blood Ca concentration are a conserved response to an activated immune response (Collage et al., 2013). Increased intracellular Ca concentration is a key feature of immune response, and several researchers have found associations between severe inflammatory responses and decreased blood Ca concentrations postpartum (Kimura et al., 2006; Bertoni et al., 2008).

Given the serious outcomes of cows experiencing unregulated inflammation and Ca dysregulation, as well as the existing relationship between the two, we must seek more consistent definitions for diagnosis of SCH and develop more robust data sets to describe acute phase proteins and cytokines in the bovine. Recently, more work has been conducted (Neves, 2023; Seminara et al., 2025) to better understand the impetus of periparturient hypocalcemia and whether it is solely driven by immune system activation (Horst et al., 2021). We need to elucidate the nature of the relationships between periparturient inflammatory status and Ca dynamics. It would be prudent to investigate how these different Ca dynamics relate to the dynamics of periparturient inflammation. Therefore, the objective of this study was to assess the patterns of select acute phase proteins, specifically haptoglobin (Hp) and serum amyloid-A (SAA), and cytokines pre- and postpartum in cows retrospectively assigned to 1 of 4 calcemia groups based on total calcium (tCa) status at 1 and 5 DIM. We hypothesized that SAA, Hp, TNF α , IFN γ , and IL-10 would be tightly associated with calcium group assignment, with greatest elevations of SAA, Hp, and proinflammatory cytokines in SCH cows postpartum and greatest elevations of the anti-inflammatory cytokine IL-10 in normal and tSCH cows.

MATERIALS AND METHODS

Study Population and Experimental Design

All procedures involving animals were approved by the Cornell University Institutional Animal Care and Use Committee, protocol no. 2019-0036. Our analysis was completed as a retrospective cohort design and included multiparous Holstein cows ($n = 96$) during the periparturient period from a previous study (Graef et al., 2025). All animals were housed between October 2019 and October 2020 in individual tiestalls at the Cornell University Ruminant Center (Harford, NY).

Feeding Management and Health Monitoring

All cows were individually fed once per day between 0700 and 0900 h for lactating cows and between 0700 and 1100 h for dry cows. Cows were all fed the same far-off dry cow and fresh cow diets after parturition. All far-off dry cow diet, close-up diet, and postpartum diet compositions were reported in Graef et al. (2025).

Cows were observed daily throughout the trial for health disorders by farm and research personnel. All postpartum cows underwent daily physical exams by the research team until 10 DIM; these included temperature, respiration rate, rumination, and metritis checks. Identification of health disorders by farm staff was as follows: clinical hypocalcemia = decline in blood Ca accompanied by clinical signs of cold extremities, trembling, and paresis; metritis = watery, fetid brown uterine discharge with or without systemic signs; retained fetal membranes = retention of fetal membranes for >24 h; mastitis = abnormal milk or an inflamed quarter; displaced abomasum = a “ping” sound upon simultaneous percussion and auscultation of the abdomen; elevated urine ketones (Ketostix, Bayer Corporation), cows with urine ketones measuring “small” or greater for 2 consecutive days were treated with 300 mL of oral propylene glycol for 5 d. No cows included in this study received supplemental Ca in the form of oral boluses, pastes, gels, or intravenous infusion; nor did they receive any antibiotics or anti-inflammatories. Body weight was measured and BCS was assigned weekly (1 to 5 scale with 0.25-point intervals; Edmonson et al., 1989) by 2 scorers and averaged for statistical analysis.

Cows were milked 3×/d at 0600, 1400, and 2200 h from calving through 9 wk postpartum. Milk weights were collected from each milking, and milk samples were collected once per week from all 3 milkings within a 24-h period. Milk sampling and analysis methods were previously described in Graef et al. (2025).

Animal Sampling and Analysis

Blood samples were collected from the coccygeal vein or artery, and methods are described in Graef et al. (2025). Prepartum samples were taken at d -28 ($-28.5 \pm \text{SD } 2.6$) and d -1 ($-1.5 \pm \text{SD } 0.5$) relative to expected calving, and postpartum samples at d 1, 2, 3, 5, and 7 relative to calving. Prepartum blood samples were collected between 0600 and 0730 h. Postpartum blood samples were collected at ~ 0700 h. All handling of blood samples and analysis for tCa, Mg, and P analysis is described in Graef et al. (2025). Pre- and postpartum serum was analyzed for Hp using a protocol described by Eckersall et al. (1999) with inter- and intra-assay CV of 8.0% and 3.0%, respectively, and SAA with a commercially available colorimetric kit (Tri-Delta Diagnostics Inc., Township, NJ), reporting intra- and interassay CV of 9.6% and 13.3%, respectively. Assays were run in duplicate (Hp) and triplicate (SAA). Spectrophotometric measurements were conducted using a tunable microplate reader (SpectraMax 190, Molecular Devices, San Jose, CA). Serum TNF α , IFN γ , and IL-10 were analyzed via a bead-based multiplex assay developed and described by Sipka et al. (2022). Intra- and interassay CV, respectively, are as follows: 4.9% and 9.0% for TNF α , 7.6% and 7.0% for IFN γ , and 6.0% and 3.0% for IL-10.

Statistical Analysis

All statistical analyses were conducted using SAS software (v. 9.4, SAS Institute Inc., Cary, NC). Descriptive statistics were generated using the FREQ and MEANS procedures. Before statistical analysis of DMI, milk yield, and blood analytes, cows were retrospectively classified into 1 of 4 calcemia groups. Other work has done this based on tCa status at 1 and 4 DIM (McArt and Neves, 2020; Seely et al., 2021; McCray et al., 2023). This retrospective cohort study did not have a blood sample from 4 DIM and, seeking to continue to prioritize the established directionality of these calcemia groupings, instead based classification on blood tCa concentrations at 1 and 5 DIM. To diagnose SCH, this study used a previously identified cutpoint of 1.95 mmol/L at 1 DIM (Seely et al., 2021), whereas for 5 DIM we identified the interquartile range of tCa concentrations and divided that range into 23 cutpoints in 0.02 mmol/L intervals. Generalized linear mixed models were created via the MIXED procedure for parity 2 and ≥ 3 , respectively, to establish milk production differences between cows above and below the tCa cutpoint being assessed (Table 1). Models included fixed effects of week of lactation, 5 DIM Ca status, and their interaction. To identify the maximum differences in milk production, final tCa cutpoints were selected. Cows below the cutpoint were

deemed SCH and produced less milk than cows above the cutpoint. This definition of diagnosis was selected based on our understanding that cows experiencing delayed or persistent SCH produce less milk than cows considered to have a normal tCa concentration at 4 DIM (Neves et al., 2018; McArt and Neves, 2020; Seely et al., 2021). The calcemia groups were as follows: NC (tCa > 1.95 mmol/L at 1 DIM and > 2.32 mmol/L at 5 DIM, $n = 53$); tSCH (tCa ≤ 1.95 mmol/L at 1 DIM and > 2.32 mmol/L at 5 DIM, $n = 15$); dSCH (tCa > 1.95 mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and pSCH (tCa ≤ 1.95 mmol at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$). A post hoc power calculation was conducted using ClinCalc (ClinCalc LLC, <https://clincalc.com>). Allowing for a 5% risk of type I error, our sample size reported 92.4% power to detect a difference of 3 kg/d (± 1.2 kg/d) in daily milk yield.

Generalized linear mixed models were created through the MIXED procedure. Pre- and postpartum samples were assessed separately and included their respective covariates. Models included the REPEATED statement to account for multiple measures to assess changes over time and detect differences between calcemia groups for blood measures, milk yield, and DMI. Seven covariance structures were tested for each model, including compound symmetry, heterogeneous compound symmetry, autoregressive order 1, heterogeneous autoregressive order 1, antedependence 1, variance components, and Toeplitz where the lowest Akaike's information criterion was selected: variance components (Littell et al., 1996). When non-normality of residual variance was evident (Hp), data were log-transformed and analysis repeated. Results are reported as least squares means or back-transformed means for transformed variables, and their respective 95% confidence intervals.

In all results, differences between means were considered statistically significant when $P \leq 0.05$ with evidence for a difference when $0.05 < P \leq 0.10$. When main effects and least squares means differed between groups, Tukey-Kramer studentized adjustments were used to account for multiple comparisons. All models included the random effect of cow nested within enrollment block and the fixed effects of calcemia group, previously assigned treatment, time, parity, and their respective 2- and 3-way interactions. Parity was defined as either second or third and greater. Variables and interactions were removed in a backward stepwise manner if $P > 0.10$. Irrespective of statistical significance, main effects of time, calcemia group, and their interaction remained in all models. Prepartum treatment assignment was not significant in any model, pre- or postpartum. Parity only remained in the model for postpartum P, milk yield (MY), ECM, and FCM. Figures and tables were created using Microsoft Excel (2018; Microsoft Corp., Redmond, WA).

Table 1. Mean milk yield for the first 9 wk of lactation at different serum total Ca (tCa) cutpoints¹ at 5 DIM in multiparous Holstein cows (n = 96) in tiestalls

tCa 5 DIM, mmol/L	% (n) ≤tCa cutpoint	% (n) >tCa cutpoint	MY ≤tCa cutpoint, kg/d	MY >tCa cutpoint, kg/d	P-value
Parity 2 (n = 59)					
1.50	0 (0)	100 (59)	—	43.5	0.01
1.80	2 (1)	98 (58)	38.7	43.5	0.01
1.85	2 (1)	98 (58)	38.7	43.5	0.01
1.90	2 (1)	98 (58)	38.7	43.5	0.01
1.95	2 (1)	98 (58)	38.7	43.5	0.01
2.00	2 (1)	98 (58)	38.7	43.5	0.01
2.05	2 (1)	98 (58)	38.7	43.5	0.01
2.10	8 (5)	92 (54)	44.2	43.4	0.44
2.15	10 (6)	90 (53)	44.6	43.3	0.08
2.20	12 (7)	88 (52)	44.6	43.3	0.06
2.22	12 (7)	88 (52)	44.6	43.3	0.06
2.24	12 (7)	88 (52)	44.6	43.3	0.06
2.26	17 (10)	83 (49)	44.1	43.3	0.17
2.28	19 (11)	83 (48)	44.3	43.2	0.08
2.30	24 (14)	76 (45)	44.3	43.2	0.08
2.32	24 (14)	76 (45)	44.4	43.1	0.02
2.34	29 (17)	71 (42)	44.5	43.0	0.002
2.36	36 (21)	64 (38)	44.4	42.9	0.003
2.38	42 (25)	58 (34)	43.9	43.1	0.09
2.40	53 (31)	47 (28)	43.9	43.1	0.09
2.45	63 (37)	37 (22)	43.7	43.0	0.14
2.50	83 (49)	17 (10)	43.7	42.7	0.06
3.00	59	0	43.5	—	0.00
Parity ≥3 (n = 37)					
1.50	3 (1)	97 (36)	44.7	46.8	0.39
1.80	8 (3)	92 (34)	44.6	46.9	0.17
1.85	8 (3)	92 (34)	44.6	46.9	0.41
1.90	8 (3)	92 (34)	44.6	46.9	0.41
1.95	8 (3)	92 (34)	44.6	46.9	0.41
2.00	8 (3)	92 (34)	44.6	46.9	0.41
2.05	11 (4)	89 (33)	45.7	46.9	0.41
2.10	19 (7)	89 (30)	46.1	46.9	0.56
2.15	22 (8)	78 (29)	45.8	47.1	0.19
2.20	27 (10)	73 (27)	46.0	47.0	0.26
2.22	27 (10)	73 (27)	46.2	47.0	0.33
2.24	30 (11)	70 (26)	45.7	47.2	0.08
2.26	30 (11)	70 (26)	45.7	47.2	0.08
2.28	32 (12)	68 (25)	45.6	47.4	0.03
2.30	38 (14)	62 (23)	45.6	47.4	0.03
2.32	38 (14)	62 (23)	45.2	47.8	0.00
2.34	41 (15)	59 (22)	46.3	47.1	0.33
2.36	46 (17)	54 (20)	46.7	46.9	0.86
2.38	49 (18)	51 (19)	46.8	46.8	0.92
2.40	57 (21)	43 (16)	46.8	46.8	0.92
2.45	65 (24)	35 (13)	46.9	46.5	0.61
2.50	78 (29)	22 (8)	47.0	46.3	0.43
3.00	100 (37)	0 (0)	46.6	49.7	0.06

¹A total of 23 cutpoints in 0.02 mmol/L intervals were identified based on the interquartile range of tCa concentrations. Models were created separately for cows in parity 2 (n = 59) and parity ≥3 (n = 37).

RESULTS

Characterization of Animals

In total, 96 multiparous Holstein cows from a previous study were included in this analysis. The final counts of cows assigned to each calcemia group based on their tCa status on 1 and 5 DIM were as follows: NC (n = 53), tSCH (n = 15), dSCH (n = 15), and pSCH (n = 13). Health events, the proportion of cows entering parity 2 versus

parity ≥3, and prepartum treatment groups are reported in Table 2. No animals died or were removed. Cows entering their second parity made up a greater proportion of the NC group, and cows entering parity ≥3 were a greater proportion of tSCH, dSCH, and pSCH groups ($P < 0.01$). Least squares means and 95% CI of BW, BCS, and previous lactation 305-d mature-equivalent MY (**ME305**) of enrolled cows by calcemia group are also reported in Table 2. No differences existed between assigned calcemia groups for previously assigned prepartum treatment

Table 2. Descriptive statistics of multiparous cows classified into 1 of 4 subclinical hypocalcemia (SCH) groups

Variable	Calcemia group ¹				P-value
	Normal (n = 53)	Transient (n = 15)	Delayed (n = 15)	Persistent (n = 13)	
Health events, n					
Clin hyp ²	1	1	0	0	—
Metritis	1	2	1	1	—
RP ³	1	0	1	0	—
DA ⁴	1	0	0	0	—
Mastitis	0	1	0	1	—
Parity, %					<0.001
2	43	4	11	3	—
≥3	13	11	4	10	—
Treatment, ⁵ %					0.93
LCFAS	16	3	3	4	—
LCPAS	10	6	4	2	—
HCFAS	14	2	5	5	—
HCPAS	16	4	3	2	—
Previous ME305, ⁶ kg	13,224 (8,111 to 17,500)	13,376 (11,096 to 16,403)	14,106 (11,132 to 18,029)	13,740 (11,692 to 17,274)	0.49
Prepartum BW, kg	736 (622, 851)	789 (594, 986)	737 (678, 822)	787 (646, 936)	0.56
Prepartum BCS, kg	3.50 (2.75, 4.0)	3.50 (3.25, 4.0)	3.50 (3.25, 3.75)	3.50 (3.25, 4.0)	0.81

¹Cows were retrospectively assigned to a calcemia group based on total Ca (tCa) concentrations at 1 and 5 DIM: normocalcemic (NC; >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM); transient SCH (tSCH: [tCa] ≤1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM); delayed SCH (dSCH: [tCa] >1.95 mmol/L at 1 DIM and ≤2.32 mmol/L at 5 DIM); and persistent SCH (pSCH: [tCa] ≤1.95 mmol/L at 1 DIM and ≤2.32 mmol/L at 5 DIM).

²Clinical hypocalcemia.

³Retained placenta.

⁴Displaced abomasum.

⁵Ingredients and analyzed composition of prepartum treatment diets are reported in previous work (Graef et al., 2025). Treatments consisted of ~0.7% Ca vs. ~1.5% Ca DM (LC vs. HC) and urine pH targeting 5.5–6.0 vs. 6.5 vs. 7.0 (FAS vs. PAS). This yielded 4 treatment groups: LCFAS, LCPAS, HCFAS, and HCPAS.

⁶Previous lactation 305-d mature-equivalent milk yield.

group ($P = 0.93$), previous ME305 ($P = 0.49$), BW ($P = 0.56$), or BCS ($P = 0.81$).

DMI and Milk Yield

Pre- and postpartum DMI are reported in Table 3 and Figure 1. Prepartum DMI was greater in tSCH cows (13.2 kg/d, 95% CI = 12.2, 14.2 kg/d) compared with NC (12.8 kg/d, 95% CI = 12.3, 13.2 kg/d) and pSCH (11.8 kg/d, 95% CI = 11.2, 12.4 kg/d) cows. Prepartum DMI decreased by day as cows approached parturition ($P < 0.001$). In the first 3 wk postpartum, DMI of cows in the pSCH group were considered lowest by SCH and time, particularly at 2, 6, 7, 15, and 19 DIM (Figure 1; $P < 0.001$). Over 9 wk, postpartum DMI steadily increased ($P < 0.001$) but otherwise did not differ by calcemia group ($P = 0.54$) or parity ($P = 0.84$), and the interaction of calcemia group and week was not statistically significant ($P = 0.33$).

Results for MY and ECM are reported in Table 3 and Figure 2. From wk 1 to 3, numerical differences in MY ($P = 0.06$) and significant differences in ECM ($P = 0.02$) indicated the greatest yields in tSCH and dSCH groups, respectively, compared with the other 3 groups. From wk 1 to 9, cows categorized as tSCH had the highest MY and

ECM, whereas pSCH cows had the lowest MY and ECM ($P < 0.001$). Cows categorized as dSCH outperformed NC and pSCH cows in MY and pSCH cows in both MY and ECM in wk 1 to 9 ($P < 0.001$).

Blood Mineral and Inflammatory Markers

Blood mineral concentrations are reported in Table 4 and Figure 3. Prepartum blood tCa, Mg, and P concentrations were not different between calcemia groups. Postpartum tCa concentrations were different by calcemia group, day, and their interactions such that the NC and dSCH groups had the greatest mean concentration of tCa at 1 DIM compared with tSCH and pSCH, but NC and tSCH had higher concentrations of tCa at 5 DIM compared with dSCH and pSCH cows ($P < 0.001$). Postpartum Mg concentrations also differed in their interaction of calcemia group and time such that Mg concentrations at 1 DIM were lowest in pSCH cows and greatest in dSCH cows (NC 1.00 mmol/L, 95% CI = 0.97, 1.04 mmol/L; vs. tSCH 1.08 mmol/L, 95% CI = 1.02, 1.15 mmol/L; vs. dSCH 0.94 mmol/L, 95% CI = 0.87, 1.01 mmol/L; vs. pSCH 0.83 mmol/L, 95% CI = 0.75, 0.90 mmol/L; $P = 0.05$).

Table 3. Least squares means and 95% CI reported by calcemia grouping of pre- (wk -3 to -1) and postpartum DMI, milk yield (MY), ECM, and FCM for multiparous Holstein cows

Variable	Calcemia group ¹				P-value		
	Normal	Transient	Delayed	Persistent	Calcemia group	Time	SCH × Time
Prepartum DMI, kg	12.8 ^a (12.3, 13.2)	13.2 ^b (12.2, 14.2)	12.5 ^{ab} (11.4, 13.6)	11.8 ^a (11.2, 12.4)	0.03	<0.001	0.56
Postpartum DMI, kg							
1 to 3 wk	18.5 (17.9, 19.1)	18.3 (17.2, 19.4)	18.5 (16.8, 20.1)	17.3 (15.3, 19.3)	0.73	<0.001	<0.001
1 to 9 wk	20.4 (19.8, 20.9)	21.0 (19.9, 22.1)	20.2 (18.9, 21.6)	19.9 (18.8, 21.0)	0.54	<0.001	0.33
Milk yield, kg							
1 to 3 wk	41.7 ^a (40.8, 42.7)	44.0 ^b (42.3, 45.6)	42.0 ^{ab} (40.4, 43.7)	40.8 ^a (38.9, 42.6)	0.06	<0.001	0.25
1 to 9 wk	44.6 ^a (44, 45.2)	47.3 ^b (46.3, 48.2)	45.8 ^c (44.9, 46.8)	43.8 ^a (42.7, 44.9)	<0.001	<0.001	0.99
ECM, kg							
1 to 3 wk	50.1 ^{ab} (48.6, 51.5)	51.1 ^a (49.2, 53)	51.7 ^a (49.8, 53.5)	48.1 ^b (46.2, 50)	0.02	0.21	0.99
1 to 9 wk	48.6 ^a (47.8, 49.4)	50.4 ^b (49.4, 51.5)	48.8 ^a (47.8, 49.8)	46.5 ^c (45.4, 47.5)	<0.001	<0.001	0.83

^{a-c}Within a row means without a common superscript differ by the main effect of calcemia group ($P \leq 0.05$).

¹Cows were retrospectively assigned to a calcemia group (SCH = subclinical hypocalcemia) based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH: $[tCa] >1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$).

Differences between calcemia groups and changes over time for Hp, SAA, and cytokines are reported in Table 5 and illustrated in Figures 4, 5, and 6, respectively. Prepartum concentrations of Hp ($P = 0.71$), SAA ($P = 0.62$), TNF α ($P = 0.35$), and IFN γ ($P = 0.60$) did not differ between calcemia groups. Prepartum concentrations of IL-10 increased numerically ($P = 0.12$) in cows classified as tSCH (23.26 pg/mL, 95% CI = 17.95, 30.14 pg/mL) compared with dSCH (18.12 pg/mL, 95% CI = 11.31, 29.03 pg/mL) and pSCH (36.49 pg/mL, 95%

CI = 11.21, 18.84). Postpartum concentrations of Hp ($P = 0.03$) and SAA ($P < 0.001$) were different by the interaction of calcemia group and time. Compared with the other 3 groups, postpartum Hp concentrations were elevated in pSCH cows (pSCH 0.96 g/mL, 95% CI = 0.40, 2.29 g/mL; vs. NC 0.12 g/mL, 95% CI = 0.10, 0.16 g/mL; vs. tSCH 0.10 g/mL, 0.04 $\pm$ 0.24 g/mL; vs. dSCH 0.21 g/mL, 0.14, 0.32 g/mL). The pSCH cows also had elevated SAA concentrations (101.14 mmol/L, 95% CI = 42.47, 24.86 mmol/L) compared with the other 3 Ca

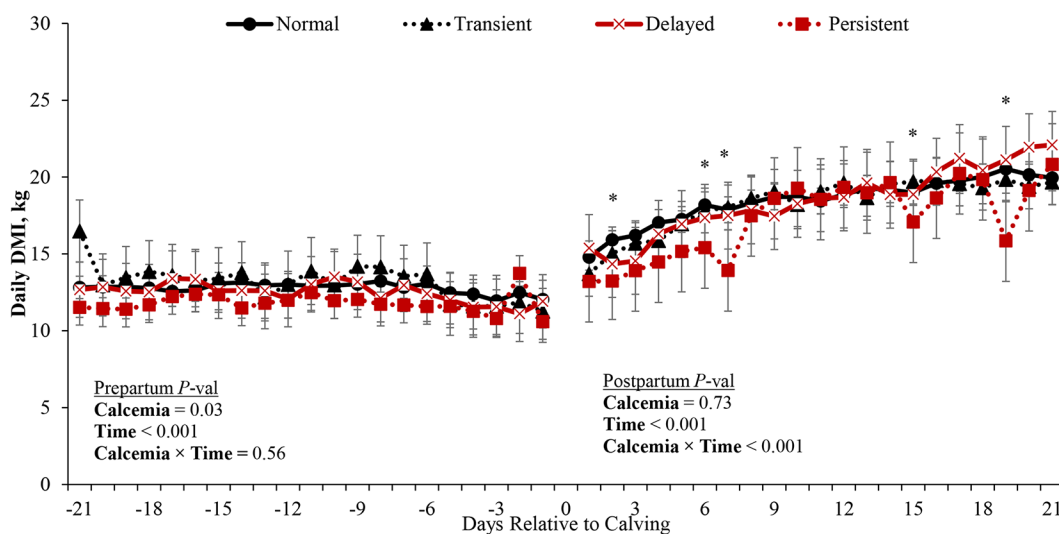


Figure 1. Least squares means and 95% CI of periparturient DMI of multiparous Holstein cows ($n = 96$) by calcemia group assignment. Cows were retrospectively assigned to a calcemia group based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH: $[tCa] >1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$). Differences in average DMI between calcemia groups within a time point, based on a Tukey adjustment $P < 0.05$, are represented with (*). SCH = subclinical hypocalcemia; P-val = P-value.

Table 4. Least squares means and 95% CI reported by subclinical hypocalcemia (SCH) grouping of pre- and postpartum DMI and blood concentrations of total Ca (tCa), P, and Mg on d -7, 1, 2, 4, and 7 relative to calving for multiparous Holstein cows retrospectively classified into 1 of 4 calcemia groups

Variable	Calcemia group ¹				P-value		
	Normal	Transient	Delayed	Persistent	Calcemia group	Time	SCH × Time
Prepartum							
tCa, mmol/L	2.40 (2.31, 2.49)	2.40 (2.23, 2.57)	2.45 (2.27, 2.62)	2.20 (2.01, 2.40)	0.27	—	—
P, mmol/L	1.97 (1.90, 2.04)	1.96 (1.83, 2.09)	2.04 (1.91, 2.17)	2.00 (1.85, 2.14)	0.77	—	—
Mg, mmol/L	0.96 (0.93, 0.98)	0.97 (0.93, 1.02)	0.92 (0.87, 0.96)	0.89 (0.84, 0.94)	0.30	—	—
Postpartum							
tCa, mmol/L	2.37 (2.34, 2.40)	2.23 (2.17, 2.29)	2.24 (2.18, 2.29)	1.98 (1.92, 2.05)	<0.001	<0.001	<0.001
P, mmol/L	1.35 (1.28, 1.42)	1.17 (1.04, 1.30)	1.36 (1.23, 1.49)	1.33 (1.04, 1.30)	0.12	<0.001	<0.001
Mg, mmol/L	0.96 (0.93, 0.98)	0.97 (0.93, 1.02)	0.92 (0.887, 0.96)	0.89 (0.84, 0.94)	0.05	0.06	0.004

¹Cows were retrospectively assigned to a calcemia group based on tCa concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, n = 53); transient SCH (tSCH: [tCa] ≤1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, n = 15); delayed SCH (dSCH: [tCa] >1.95 mmol/L at 1 DIM and ≤2.32 mmol/L at 5 DIM, n = 15); and persistent SCH (pSCH: [tCa] ≤1.95 mmol/L at 1 DIM and ≤2.32 mmol/L at 5 DIM, n = 13).

groups. Postpartum concentrations of Hp ($P = 0.03$) and SAA ($P < 0.001$) differed by the interaction of calcemia group by time (Figures 4 and 5). Concentrations of Hp in pSCH (0.66 g/mL, 95% CI = 0.21, 2.04 g/mL) cows were elevated compared with NC, tSCH, and dSCH cows as early as 1 DIM (NC 0.16 g/mL, 95% CI = 0.12, 0.22 g/mL; tSCH 0.12 g/mL, 95% CI = 0.04, 0.36 g/mL; dSCH 0.16 g/mL, 95% CI = 0.10, 0.28 g/mL). Relative to NC and tSCH cows, which experienced a steady decline in SAA concentrations following parturition, pSCH cows experienced peak concentrations of SAA at 5 DIM followed by a decline thereafter ($P \leq 0.001$; Figure 5). Although dSCH cows had numerical elevations of circulating SAA at 5 and 7 DIM compared with

NC cows ($P = 0.07$), at 1 DIM cows assigned to NC and tSCH groups had numerically higher SAA compared with dSCH cows ($P = 0.06$). Postpartum concentrations of TNF α , IFN γ , and IL-10 were not different by calcemia group, time, or the interaction of calcemia group and time.

DISCUSSION

The objective of this study was to assess the association between calcium dynamics and patterns of Hp, SAA, and cytokines during the periparturient period as well as cow performance in multiparous dairy cows. In our cur-

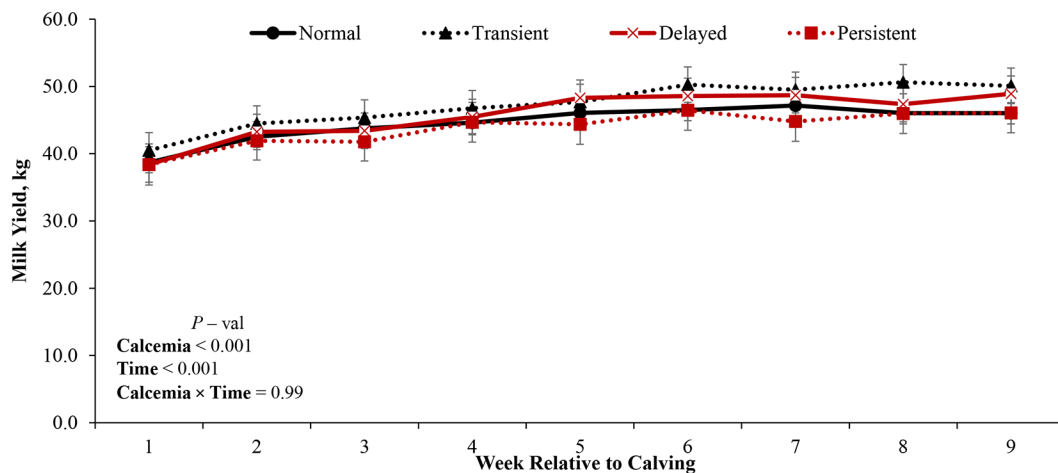


Figure 2. Least squares means and 95% CI of milk yield of multiparous Holstein cows (n = 96) by calcemia group assignment. Cows were retrospectively assigned to a calcemia group based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, n = 53); transient SCH (tSCH: [tCa] ≤1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, n = 15); delayed SCH (dSCH: [tCa] >1.95 mmol/L at 1 DIM and ≤2.32 mmol/L at 5 DIM, n = 15); and persistent SCH (pSCH: [tCa] ≤1.95 mmol/L at 1 DIM and ≤2.32 mmol/L at 5 DIM, n = 13). The final model included calcemia group, time, parity, and calcemia group × time. SCH = subclinical hypocalcemia; P-val = P-value.

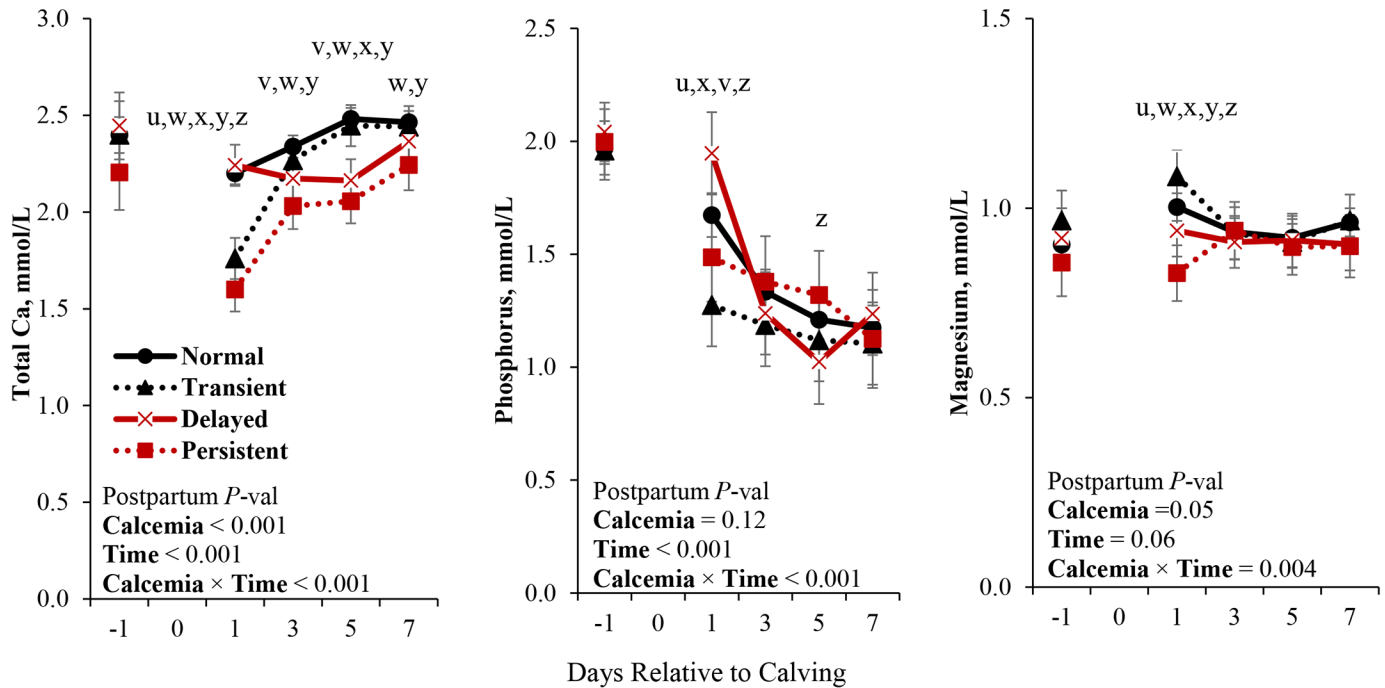


Figure 3. Least squares means and 95% CI of periparturient serum mineral concentrations of total Ca (a), phosphorus (b), and magnesium (c) in multiparous Holstein cows ($n = 96$) by calcemia group. Cows were retrospectively assigned to a calcemia group based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH: $[tCa] >1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$). The final model for total Ca, phosphorus, and magnesium prepartum included calcemia group, and the final model for postpartum minerals included calcemia group, time, and calcemia group $\times$ time. Differences in average blood mineral concentrations among calcemia groups within a time point, based on a Tukey adjustment $P < 0.05$, are represented with letters as follows: u = NC vs. tSCH, v = NC vs. dSCH, w = NC vs. pSCH, x = tSCH vs. dSCH, y = tSCH vs. pSCH, z = pSCH vs. dSCH. SCH = subclinical hypocalcemia; P -val = P -value.

Table 5. Least squares means and 95% CI reported by calcemia grouping (SCH = subclinical hypocalcemia) of pre- and postpartum blood concentrations of haptoglobin (Hp), SAA, TNF α , IFN γ , and IL-10 for multiparous Holstein cows; prepartum blood samples are a single time point d -1 relative to expected calving, and postpartum samples are reported LSM of 1, 3, 5, and 7 DIM

Variable	Calcemia group ¹				P -value		
	Normal	Transient	Delayed	Persistent	Calcemia group	Time	Calcemia $\times$ Time
Prepartum							
Hp, g/mL	0.06 (0.05, 0.09)	0.05 (0.02, 0.16)	0.06 (0.04, 0.11)	0.13 (0.04, 0.48)	0.71	—	—
SAA, mmol/L	13.62 (10.43, 17.8)	16.84 (5.65, 50.21)	14.21 (8.72, 23.15)	6.19 (1.82, 21.06)	0.62	—	—
TNF α , pg/mL	7.04 (4.07, 12.19)	37.82 (3.29, 434.76)	4.02 (1.38, 11.68)	21.1 (1.36, 326.65)	0.35	—	—
IFN γ , pg/mL	11.29 (8.47, 15.04)	21.61 (6.6, 70.76)	10.56 (6.23, 17.91)	20.22 (5.35, 76.42)	0.60	—	—
IL-10, pg/mL	23.26 (17.95, 30.14)	73.08 (25.48, 209.64)	18.12 (11.31, 29.03)	36.49 (11.21, 118.84)	0.12	—	—
Postpartum							
Hp, g/mL	0.12 (0.1, 0.16)	0.10 (0.04, 0.24)	0.21 (0.14, 0.32)	0.96 (0.4, 2.29)	<0.001	0.02	0.03
SAA, mmol/L	33.05 (25.89, 42.19)	56.33 (23.7, 133.89)	38.14 (25.86, 58.05)	101.14 (42.47, 240.86)	0.08	<0.001	<0.001
TNF α , pg/mL	4.23 (2.48, 7.24)	11.64 (1.75, 77.6)	7.24 (2.98, 17.63)	13.79 (2.04, 93.34)	0.42	0.06	0.66
IFN γ , pg/mL	8.18 (5.53, 12.08)	20.07 (5.03, 80.02)	8.02 (4.2, 15.31)	10.91 (2.72, 43.75)	0.64	0.19	0.51
IL-10, pg/mL	16.69 (13.15, 21.18)	29.23 (12.46, 68.55)	17.46 (11.74, 25.97)	18.42 (7.83, 43.33)	0.66	0.16	0.99

¹Cows were retrospectively assigned to a calcemia group based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH: $[tCa] >1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$).

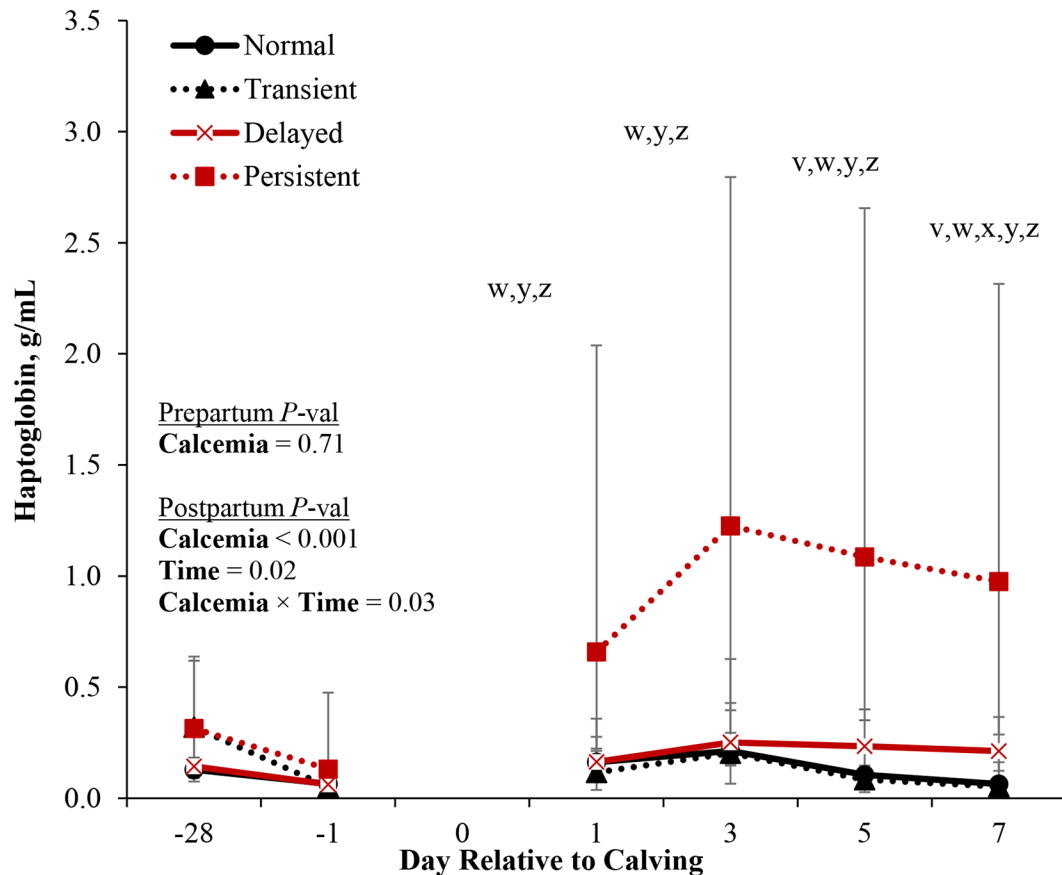


Figure 4. Least squares means and 95% CI of periparturient serum haptoglobin (Hp) in multiparous Holstein cows ($n = 96$) by calcemia group assignment. Cows were retrospectively assigned to a calcemia group based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH: [tCa] ≤ 1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH: [tCa] >1.95 mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH: [tCa] ≤ 1.95 mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$). The final model for Hp prepartum included calcemia group; the postpartum model included calcemia group, time, and calcemia group $\times$ time. Differences in average Hp concentrations among calcemia groups within a time point, based on a Tukey adjustment $P < 0.05$, are represented with letters as follows: v = NC vs. dSCH, w = NC vs. pSCH, x = tSCH vs. dSCH, y = tSCH vs. pSCH, z = pSCH vs. dSCH. SCH = subclinical hypocalcemia; P -val = P -value.

rent study, prepartum intake was increased in tSCH cows compared with NC cows. Other work diagnosing SCH at 24 h (similar to tSCH cows), such as that conducted by Jawor et al. (2012), found that these cows (24-h tCa ≤ 1.8 mmol/L) had greater pre- and postpartum intakes compared with eucalcemic cows at 24 h postpartum. Opgenorth et al. (2024), conducting intramammary LPS infusions, reported increased IL-10 and increased DMI in early-lactation animals. This is contrary to the long-standing understanding that declines in prepartum DMI are associated with impaired immune function and inflammatory stimuli; however, the order of events remains unclear (Drackley, 1999). Another retrospective analysis by Seely et al. (2021) similarly found little relationship between calcemia group or postpartum SCH dynamics and prepartum DMI, apart from a general decrease of intakes as all cows approached calving. Goff et al. (2020) explored the association between Ca concentra-

tion, DMI, and rumination time, and observed reductions in periparturient rumination time in cows with low circulating postpartum Ca compared with those with high Ca. More recently, the association between prepartum rumination and change in DMI as possible indicators of postparturient Ca status has been explored (Emam et al., 2023). Similarly, Emam et al. (2023) found little evidence of association between prepartum rumination time or DMI with blood mineral status at 3 DIM. It is possible that with a larger data set that is more consistent in the time points used to identify SCH cows, we may be able to elucidate more differences in prepartum DMI between acute decreases in blood Ca immediately postpartum versus cows suffering from prolonged low circulating tCa. In this study, differences in postpartum DMI began as early as 2 DIM. Interestingly, differences in postpartum Hp, SAA, and proinflammatory cytokines in dSCH and pSCH groups were observed by 3 DIM. For

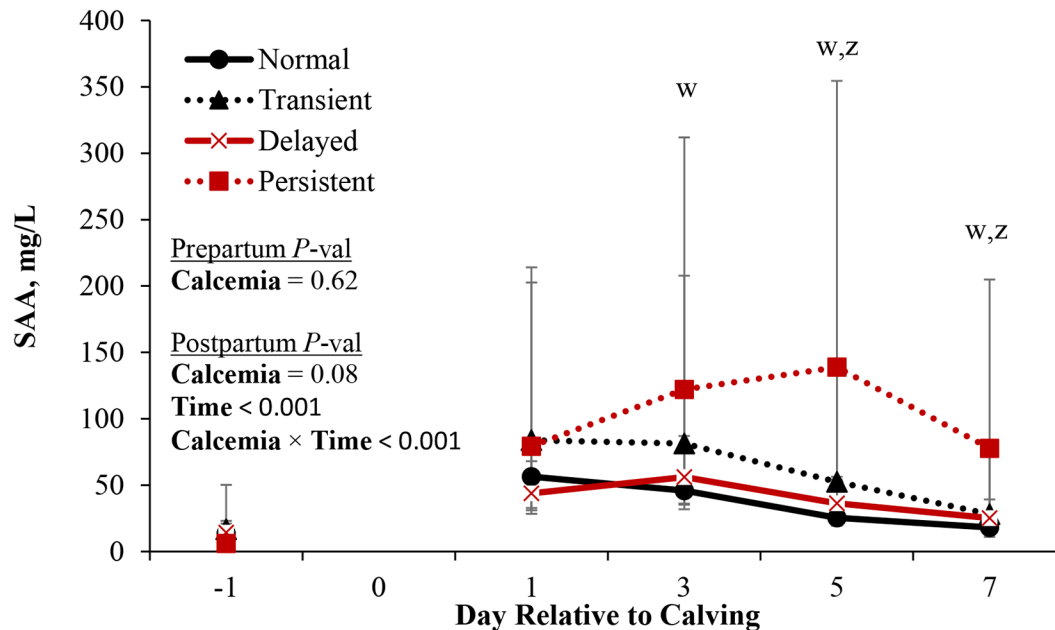


Figure 5. Least squares means and 95% CI of periparturient SAA concentrations in multiparous Holstein cows ($n = 96$) by calcemia group assignment. Cows were retrospectively assigned to a calcemia group based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH: $[tCa] >1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$). The final model SAA prepartum included calcemia group; the postpartum model included calcemia group, time, and SCH $\times$ time. Differences in average SAA concentrations among calcemia groups within a time point, based on a Tukey adjustment $P < 0.05$, are represented with letters as follows: w = NC vs. pSCH, z = pSCH vs. dSCH. SCH = subclinical hypocalcemia; P -val = P -value.

the first 3 wk postpartum, DMI was significantly lower in the pSCH cows compared with the other 3 calcemia groups; however, when DMI was evaluated over the first 9 wk postpartum, we failed to detect a difference in DMI between calcemia groups. It might be postulated that declines in DMI in the early postpartum period (wk 1 to 3) may at least in part be driven by these increases in inflammatory indicators. We know that inflammation in the early postpartum period is associated with early declines in postpartum DMI (McCarthy et al., 2016). Similarly, others have also reported reductions in DMI and rumination during the early lactation period in cows diagnosed with SCH (Goff et al., 2020; Seely et al., 2021).

Milk yield and ECM were affected by calcium CH grouping. Milk production was not different between cows categorized as tSCH and dSCH, and both outperformed NC cows. This has been similarly reported in other studies comparing SCH and NC based on a single time point for a diagnosis <24 h postpartum (Jawor et al., 2012). Other researchers have reported poorer production outcomes in cows classified as dSCH compared with NC or tSCH cows, which contradicts our findings (Neves et al., 2018; McArt and Neves, 2020; Seely et al., 2021). Some of the discrepancy between this study and previous could be 3-fold. First, treatment diets in the prepartum period used anionic salts and varied levels of Ca, which

may have influenced MY in calcemia groups. Second, 2 of the previous studies (Neves et al., 2018; McArt and Neves, 2020) were conducted in commercial freestall herds, whereas this current study and the study by Seely et al. (2021) were based on data from cows in tiestalls at the Cornell University Ruminant Center (Harford, NY); it is possible that housing animals in tiestalls, variations in other management, and health event diagnosis account in part for this discrepancy. Alternatively, some variation has been reported in cutpoints for SCH at 1 DIM from other studies using this classification system, raising the possibility that some cows in the present study may have been misclassified. Furthermore, with a larger sample size, we may improve our ability to confirm the discrepancy in performance of dSCH cows, given the low number of animals ($n = 15$) in the present study, especially when divided by parity groups. Finally, it is prudent to recall that prepartum treatment diets fed in this study may have had undue influence on performance parameters, irrespective of SCH classification.

The differences observed in tCa concentrations by calcemia group and the interaction of calcemia group and time were expected due to the methods used to retrospectively assign cows to calcemia groups for this study. A relationship exists between Ca and P regulation, particularly around the transition period, and is mediated

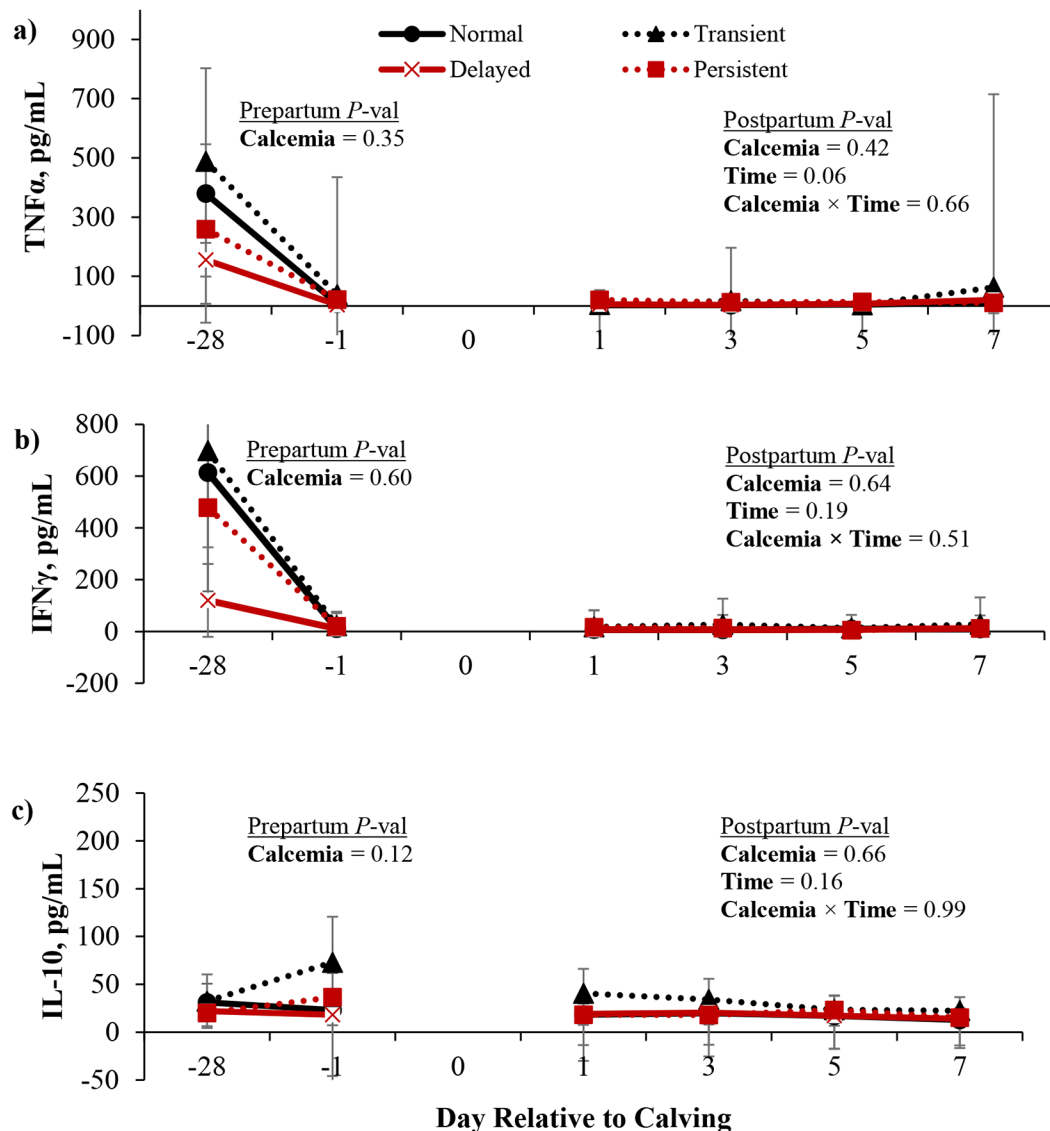


Figure 6. Least squares means and 95% CI of periparturient serum TNF- α (a), serum IFN γ (b), and IL-10 (c) in multiparous Holstein cows ($n = 96$) by calce mia group assignment. Cows were retrospectively assigned to a calce mia group based on total Ca (tCa) concentrations at 1 and 4 DIM: normocalcemic (NC: >1.95 mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 53$); transient SCH (tSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and >2.32 mmol/L at 5 DIM, $n = 15$); delayed SCH (dSCH: $[tCa] >1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 15$); and persistent SCH (pSCH: $[tCa] \leq 1.95$ mmol/L at 1 DIM and ≤ 2.32 mmol/L at 5 DIM, $n = 13$). The final model for TNF α , IFN γ , and IL-10 prepartum included calce mia grouping; the postpartum model included calce mia group, time, and calce mia group $\times$ time. SCH = subclinical hypocalcemia; P -val = P -value.

by parathyroid hormone activity on receptors on bone (Horst, 1986). Milk production elicits a large demand for P as well as Ca, often resulting in a decline in circulating P and Ca (Littledike, 1976). Although exploring P homeostasis was not the main objective of our study, our results appear to follow this trend. With the understanding that blood concentrations of Mg serve an important role in mineral transport and metabolism, particularly for Ca homeostasis, the temporarily lower circulating concentrations of Mg found in pSCH cows may support our understanding of the role of Mg in modulating Ca

metabolism, particularly during the first day of lactation (Goff, 2006). Other studies have similarly evaluated Mg in relation to SCH animals and have found that SCH animals generally have small decreases in blood Mg (Martinez et al., 2012). Although statistically significant, the numerical differences are relatively small and are not considered of biological relevance.

In this study, prepartum Hp and SAA concentrations were not different between calce mia groups. Although most cytokine results were not remarkable prepartum, the tendency for an elevation of IL-10, an anti-inflamma-

tory cytokine, in cows subsequently categorized as tSCH immediately before calving is noteworthy. This supports the working hypothesis that, through the normal periparturient inflammatory processes, the capacity for cows to limit proinflammatory stimulation appropriately may be a key factor in a successful transition (Bradford et al., 2015; Trevisi et al., 2015).

Although nonspecific, circulating Hp and SAA are relatively well-defined markers of inflammation in the bovine and are frequently used as indicators of systemic inflammation in dairy cows (LeBlanc, 2023). In this study, postpartum Hp concentrations in pSCH cows increased from 1 to 7 DIM, whereas NC, tSCH, and dSCH cows generally decreased over time. A large epidemiological study conducted in New York in 2015 included over 1,400 cows across 72 farms and reported that cows with blood Hp concentrations ≥ 0.45 g/L were 4.2 times more likely to be culled in the first 30 DIM (Kerwin et al., 2022). In the current study, the NC and tSCH cows never reached this threshold; however, cows categorized as pSCH met and exceeded this threshold by 1 DIM and remained elevated through 7 DIM. This alone is not indicative of a causal relationship between SCH status and inflammatory dynamics, but a disproportionate number of animals in the pSCH group experienced elevations in Hp concentrations that have been shown to increase risk of culling, and this adds to our current understanding of this association. This relationship is further supported by the rapid increase in SAA concentrations in pSCH cows, particularly at 3 DIM. Several studies exploring the relationships between protein expression of SAA and subclinical hypocalcemia also observed rapid increases in Hp and SAA in cows diagnosed with SCH on the day of calving (Wang et al., 2016; Fan et al., 2017).

Although dSCH and pSCH cows have been considered together as dyscalcemic due to poor health and reduced reproductive and productive outcomes (Neves et al., 2018; Seely and McArt, 2022; McCray et al., 2023), it is noteworthy in this data set that when inflammation is considered, pSCH cows are meaningfully different. Risk of pathologic increases in systemic inflammation, hyperketonemia, and elevated nonesterified fatty acids have long been thought to be predicated by prolonged dysregulation of Ca at the end of gestation and onset of lactation (Dohoo and Martin, 1984; Curtis et al., 1985; Goff et al., 1991; Caixeta et al., 2017). Several studies have demonstrated that rapid depletions in circulating Ca can be induced following intravenous LPS challenges (Waldron et al., 2003; Horst et al., 2020; Chandler et al., 2022). Although the cause is not well understood, measurable decreases in blood Ca concentrations in the face of immune activation are thought to be a protective response in other mammalian species (Malcolm et al., 1989; Zaloga et al., 1992). It has been suggested that

unresolved or chronic inflammation is the driving factor behind subclinical and clinical decreases in circulating Ca and promotes increased risk of other periparturient dysregulation (Horst et al., 2021). Results from our current study sampling time points necessitated a selection of an appropriate blood Ca cutpoint for d 5 that would best represent the directionality of subclinical declines of blood Ca in delayed and persistent SCH cows; establishing this based on MY of the enrolled cows may suggest some circular reasoning, and it is important to remember that an array of factors may directly or indirectly “cause” declines in milk production. This work only allows us to draw conclusions of associations between inflammation and SCH, but is not adequate to propose a causal relationship in either direction. However, our findings suggest that the relationship between inflammation and Ca dynamics during the periparturient period may be more complex than previously hypothesized.

CONCLUSIONS

Our results suggest that SCH dynamics are associated with inflammatory markers during the periparturient period. Blood mineral status, DMI, and milk production were all affected by SCH dynamics and grouping, in the first 7 DIM, 3 wk, and 9 wk of lactation, respectively. Distinct differences in inflammatory markers and performance metrics, particularly of dSCH cows compared with pSCH cows, suggest that more work needs to be done to confirm diagnostic thresholds of tCa at 1 and 5 DIM and to ensure that cows can be consistently classified correctly in a wide variety of dairy system types. Additional work, including more frequent and further-reaching time points, should be done to assess the nature of the relationship between inflammatory marker and SCH dynamics.

NOTES

The authors thank the staff at the Cornell University Ruminant Center (Harford, NY) for their care of the animals and extend their appreciation to Charlene Ryan, Lisa Furman, Liza Ferro, and the many undergraduate student researchers (Cornell University, Ithaca, NY) for their invaluable assistance with sample collection in the study, contributing to this analysis. This research was supported in part by the intramural research program of the US Department of Agriculture, National Institute of Food and Agriculture, Hatch Project NYC-127955. The findings and conclusions in this publication have not been formally disseminated by the US Department of Agriculture and should not be construed to represent any agency determination or policy. All procedures involving animals were approved by the Cornell University Insti-

tutional Animal Care and Use Committee, protocol no. 2019-0036. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: dSCH = delayed SCH; Hp = haptoglobin; ME305 = 305-d mature-equivalent MY; MY = milk yield; NC = normocalcemia; pSCH = persistent SCH; SCH = subclinical hypocalcemia; tCa = total calcium; tSCH = transient SCH.

REFERENCES

- Bauman, D. E., and W. B. Currie. 1980. Partitioning of nutrients during pregnancy and lactation: A review of mechanisms involving homeostasis and homeorhesis. *J. Dairy Sci.* 63:1514–1529. [https://doi.org/10.3168/jds.S0022-0302\(80\)83111-0](https://doi.org/10.3168/jds.S0022-0302(80)83111-0).
- Bertoni, G., A. Minuti, and E. Trevisi. 2015. Immune system, inflammation and nutrition in dairy cattle. *Anim. Prod. Sci.* 55:943. <https://doi.org/10.1071/AN14863>.
- Bertoni, G., E. Trevisi, X. Han, and M. Bionaz. 2008. Effects of inflammatory conditions on liver activity in puerperium period and consequences for performance in dairy cows. *J. Dairy Sci.* 91:3300–3310. <https://doi.org/10.3168/jds.2008-0995>.
- Bradford, B. J., and T. H. Swartz. 2020. Review: Following the smoke signals: inflammatory signaling in metabolic homeostasis and homeorhesis in dairy cattle. *Animal* 14(Suppl. 1):s144–s154. <https://doi.org/10.1017/S1751731119003203>.
- Bradford, B. J., K. Yuan, J. K. Farney, L. K. Mamedova, and A. J. Carpenter. 2015. Invited review: Inflammation during the transition to lactation: New adventures with an old flame. *J. Dairy Sci.* 98:6631–6650. <https://doi.org/10.3168/jds.2015-9683>.
- Caixeta, L. S., P. A. Ospina, M. B. Capel, and D. V. Nysdam. 2017. Association between subclinical hypocalcemia in the first 3 days of lactation and reproductive performance of dairy cows. *Theriogenology* 94:1–7. <https://doi.org/10.1016/j.theriogenology.2017.01.039>.
- Chandler, T., T. Westhoff, A. Sipka, T. Overton, and S. Mann. 2022. Lipopolysaccharide challenge following intravenous amino acid infusion in postpartum dairy cows: II. Clinical and inflammatory responses. *J. Dairy Sci.* 105:4611–4623. <https://doi.org/10.3168/jds.2021-12127>.
- Collage, R. D., G. M. Howell, X. Zhang, J. L. Stripay, J. S. Lee, D. C. Angus, and M. R. Rosengart. 2013. Calcium supplementation during sepsis exacerbates organ failure and mortality via calcium/calmodulin-dependent protein kinase signaling. *Crit. Care Med.* 41:e352–e360. <https://doi.org/10.1097/CCM.0b013e31828cf436>.
- Curtis, C. R., H. N. Erb, C. J. Sniffen, R. D. Smith, and D. S. Kronfeld. 1985. Path analysis of dry period nutrition, postpartum metabolic and reproductive disorders, and mastitis in Holstein cows. *J. Dairy Sci.* 68:2347–2360. [https://doi.org/10.3168/jds.S0022-0302\(85\)81109-7](https://doi.org/10.3168/jds.S0022-0302(85)81109-7).
- Dohoo, I. R., and S. Martin. 1984. Subclinical ketosis: Prevalence and associations with production and disease. *Can. J. Comp. Med.* 48:1–5.
- Drackley, J. K. 1999. Biology of dairy cows during the transition period: The final frontier? *J. Dairy Sci.* 82:2259–2273. [https://doi.org/10.3168/jds.S0022-0302\(99\)75474-3](https://doi.org/10.3168/jds.S0022-0302(99)75474-3).
- Drackley, J. K., H. M. Dann, N. Douglas, N. A. J. Guretzky, N. B. Litherland, J. P. Underwood, and J. J. Loo. 2005. Physiological and pathological adaptations in dairy cows that may increase susceptibility to periparturient diseases and disorders. *Ital. J. Anim. Sci.* 4:323–344. <https://doi.org/10.4081/ijas.2005.323>.
- Eckersall, P. D., J. Rodgers, M. Murray, and P. G. Kennedy. 1999. Haptoglobin, the acute phase response and natural human immunity to trypanosomes. *Parasitol. Today* 15:251–252. [https://doi.org/10.1016/S0169-4758\(99\)01459-3](https://doi.org/10.1016/S0169-4758(99)01459-3).
- Edmonson, A. J., I. J. Lean, L. D. Weaver, T. Farver, and G. Webster. 1989. A body condition scoring chart for Holstein dairy cows. *J. Dairy Sci.* 72:68–78. [https://doi.org/10.3168/jds.S0022-0302\(89\)79081-0](https://doi.org/10.3168/jds.S0022-0302(89)79081-0).
- Emam, M. H., E. Shepley, M. M. Mahmoud, M. Ruch, S. Elmaghawry, W. Abdelrazik, A. M. Abdelaal, B. A. Crooker, and L. S. Caixeta. 2023. The association between prepartum rumination time, activity and dry matter intake and subclinical hypocalcemia and hypomagnesemia in the first 3 days postpartum in Holstein dairy cows. *Animals (Basel)* 13:1621. <https://doi.org/10.3390/ani13101621>.
- Fan, Z., S. Shu, C. Xu, X. Xiao, G. Wang, Y. Bai, C. Xia, L. Wu, H. Zhang, C. Xu, and W. Yang. 2017. Protein profiling of plasma proteins in dairy cows with subclinical hypocalcaemia. *Ir. Vet. J.* 70:3. <https://doi.org/10.1186/s13620-017-0082-0>.
- Goff, J. P. 2006. Macromineral physiology and application to the feeding of the dairy cow for prevention of milk fever and other periparturient mineral disorders. *Anim. Feed Sci. Technol.* 126:237–257. <https://doi.org/10.1016/j.anifeeds.2005.08.005>.
- Goff, J. P., A. Hohman, and L. L. Timms. 2020. Effect of subclinical and clinical hypocalcemia and dietary cation-anion difference on rumination activity in periparturient dairy cows. *J. Dairy Sci.* 103:2591–2601. <https://doi.org/10.3168/jds.2019-17581>.
- Goff, J. P., A. Liesegang, and R. L. Horst. 2014. Diet-induced pseudohypoparathyroidism: A hypocalcemia and milk fever risk factor. *J. Dairy Sci.* 97:1520–1528. <https://doi.org/10.3168/jds.2013-7467>.
- Goff, J. P., T. A. Reinhardt, and R. L. Horst. 1991. Enzymes and factors controlling vitamin D metabolism and action in normal and milk fever cows. *J. Dairy Sci.* 74:4022–4032. [https://doi.org/10.3168/jds.S0022-0302\(91\)78597-4](https://doi.org/10.3168/jds.S0022-0302(91)78597-4).
- Graef, G., A. Kerwin, L. Ferro, S. Ordaz-Puga, C. Ryan, T. Westhoff, D. Barbano, K. Glosson, K. Zanzalari, J. Chapman, and T. R. Overton. 2025. Effect of prepartum dietary cation-anion difference strategy and level of dietary calcium on postpartum blood calcium status and milk production of multiparous Holstein cows. *J. Dairy Sci.* <https://doi.org/10.3168/jds.2024-25783>.
- Horst, E. A., S. K. Kvidera, and L. H. Baumgard. 2021. Invited review: The influence of immune activation on transition cow health and performance—A critical evaluation of traditional dogmas. *J. Dairy Sci.* 104:8380–8410. <https://doi.org/10.3168/jds.2021-20330>.
- Horst, E. A., E. J. Mayorga, M. Al-Qaisi, M. A. Abeyta, S. L. Portner, C. S. McCarthy, B. M. Goetz, S. K. Kvidera, and L. H. Baumgard. 2020. Effects of maintaining eucalcemia following immunoactivation in lactating Holstein dairy cows. *J. Dairy Sci.* 103:7472–7486. <https://doi.org/10.3168/jds.2020-18268>.
- Horst, R. L. 1986. Regulation of calcium and phosphorus homeostasis in the dairy cow. *J. Dairy Sci.* 69:604–616. [https://doi.org/10.3168/jds.S0022-0302\(86\)80445-3](https://doi.org/10.3168/jds.S0022-0302(86)80445-3).
- Jawor, P. E., J. M. Huzzey, S. J. LeBlanc, and M. A. von Keyserlingk. 2012. Associations of subclinical hypocalcemia at calving with milk yield, and feeding, drinking, and standing behaviors around parturition in Holstein cows. *J. Dairy Sci.* 95:1240–1248. <https://doi.org/10.3168/jds.2011-4586>.
- Kerwin, A. L., W. S. Burhans, S. Mann, D. V. Nysdam, S. K. Wall, K. M. Schoenberg, K. L. Perfield, and T. R. Overton. 2022. Transition cow nutrition and management strategies of dairy herds in the northeastern United States: Part II—Associations of metabolic- and inflammation-related analytes with health, milk yield, and reproduction. *J. Dairy Sci.* 105:5349–5369. <https://doi.org/10.3168/jds.2021-20863>.
- Kimura, K., T. Reinhardt, and J. Goff. 2006. Parturition and hypocalcemia blunts calcium signals in immune cells of dairy cattle. *J. Dairy Sci.* 89:2588–2595. [https://doi.org/10.3168/jds.S0022-0302\(06\)72335-9](https://doi.org/10.3168/jds.S0022-0302(06)72335-9).
- LeBlanc, S. J. 2023. Relationship of peripartum inflammation with reproductive health in dairy cows. *JDS Commun.* 4:230–234. <https://doi.org/10.3168/jdsc.2022-0328>.
- Littell, R., G. Milliken, W. Stroup, and R. Wolfinger. 1996. SAS System for Mixed Models. SAS Institute Inc., Cary, NC.
- Littledike, E. 1976. Relationship of milk secretion to hypocalcemia in the dairy cow. *J. Dairy Sci.* 59:1947–1953. [https://doi.org/10.3168/jds.S0022-0302\(76\)84466-9](https://doi.org/10.3168/jds.S0022-0302(76)84466-9).
- Malcolm, D. S., G. P. Zaloga, and J. W. Holaday. 1989. Calcium administration increases the mortality of endotoxin shock in rats. *Crit. Care Med.* 17:900–903. <https://doi.org/10.1097/00003246-198909000-00012>.
- Martinez, N., C. A. Risco, F. S. Lima, R. S. Bisinotto, L. F. Greco, E. S. Ribeiro, F. Maunsell, K. Galvao, and J. E. Santos. 2012. Evaluation of periparturient calcium status, energetic profile, and neutrophil func-

- tion in dairy cows at low or high risk of developing uterine disease. *J. Dairy Sci.* 95:7158–7172. <https://doi.org/10.3168/jds.2012-5812>.
- McArt, J. A. A., and R. C. Neves. 2020. Association of transient, persistent, or delayed subclinical hypocalcemia with early lactation disease, removal, and milk yield in Holstein cows. *J. Dairy Sci.* 103:690–701. <https://doi.org/10.3168/jds.2019-17191>.
- McCarthy, M., T. Yasui, M. Felipe, and T. Overton. 2016. Associations between the degree of early lactation inflammation and performance, metabolism, and immune function in dairy cows. *J. Dairy Sci.* 99:680–700. <https://doi.org/10.3168/jds.2015-9694>.
- McCrack, H. A., C. R. Seely, and J. A. A. McArt. 2023. Does knowledge of blood calcium concentration at 2 days postpartum affect decisions of calcium supplementation? *JDS Commun.* 5:200–204. <https://doi.org/10.3168/jdsc.2023-0456>.
- Neves, R. C. 2023. Relationship between calcium dynamics and inflammatory status in the transition period of dairy cows. *JDS Commun.* 4:225–229. <https://doi.org/10.3168/jdsc.2022-0348>.
- Neves, R. C., B. M. Leno, K. D. Bach, and J. A. A. McArt. 2018. Epidemiology of subclinical hypocalcemia in early-lactation Holstein dairy cows: The temporal associations of plasma calcium concentration in the first 4 days in milk with disease and milk production. *J. Dairy Sci.* 101:9321–9331. <https://doi.org/10.3168/jds.2018-14587>.
- Opgenorth, J., E. J. Mayorga, M. A. Abeyta, B. M. Goetz, S. Rodriguez-Jimenez, A. D. Freestone, J. L. McGill, and L. H. Baumgard. 2024. Intravenous lipopolysaccharide challenge in early versus mid-lactation dairy cattle. I: The immune and inflammatory responses. *J. Dairy Sci.* 107:6225–6239. <https://doi.org/10.3168/jds.2023-24350>.
- Reinhardt, T. A., J. D. Lippolis, B. J. McCluskey, J. P. Goff, and R. L. Horst. 2011. Prevalence of subclinical hypocalcemia in dairy herds. *Vet. J.* 188:122–124. <https://doi.org/10.1016/j.tvjl.2010.03.025>.
- Seely, C., and J. McArt. 2022. The association of subclinical hypocalcemia at 4 days in milk with reproductive outcomes in multiparous Holstein cows. *JDS Commun.* 4:111–115. <https://doi.org/10.3168/jdsc.2022-0279>.
- Seely, C. R., B. M. Leno, A. L. Kerwin, T. R. Overton, and J. A. A. McArt. 2021. Association of subclinical hypocalcemia dynamics with dry matter intake, milk yield, and blood minerals during the periparturient period. *J. Dairy Sci.* 104:4692–4702. <https://doi.org/10.3168/jds.2020-19344>.
- Seminara, J., C. Seely, and J. McArt. 2025. Acute phase responses in clinically healthy multiparous Holsteins with and without calcium dysregulation during the early postpartum period. *J. Dairy Sci.* 108:1930–1939. <https://doi.org/10.3168/jds.2024-25300>.
- Sipka, A., S. Mann, S. Babasyan, H. Freer, and B. Wagner. 2022. Development of a bead-based multiplex assay to quantify bovine interleukin-10, tumor necrosis factor- α , and interferon- γ concentrations in plasma and cell culture supernatant. *JDS Commun.* 3:207–211. <https://doi.org/10.3168/jdsc.2021-0191>.
- Sordillo, L. M., and S. L. Aitken. 2009. Impact of oxidative stress on the health and immune function of dairy cattle. *Vet. Immunol. Immunopathol.* 128:104–109. <https://doi.org/10.1016/j.vetimm.2008.10.305>.
- Trevisi, E., N. Jahan, G. Bertoni, A. Ferrari, and A. Minuti. 2015. Pro-inflammatory cytokine profile in dairy cows: Consequences for new lactation. *Ital. J. Anim. Sci.* 14:3862. <https://doi.org/10.4081/ijas.2015.3862>.
- Waldron, M., B. Nonnecke, T. Nishida, R. Horst, and T. Overton. 2003. Effect of lipopolysaccharide infusion on serum macromineral and vitamin D concentrations in dairy cows. *J. Dairy Sci.* 86:3440–3446. [https://doi.org/10.3168/jds.S0022-0302\(03\)73948-4](https://doi.org/10.3168/jds.S0022-0302(03)73948-4).
- Wang, P. X., S. Shu, C. Xia, Z. Wang, L. Wu, B. Wang, C. C. Xu, and J. Liu. 2016. Protein expression in dairy cows with and without subclinical hypocalcaemia. *N. Z. Vet. J.* 64:101–106. <https://doi.org/10.1080/00480169.2015.1100970>.
- Zaloga, G., A. Sager, K. Black, and R. Prielipp. 1992. Low dose calcium administration increases mortality during septic peritonitis in rats. *Circ. Shock* 37:226–229.

ORCIDS

- G. M. Graef, <https://orcid.org/0000-0002-0163-8845>
 S. Tompkins, <https://orcid.org/0009-0006-5349-9265>
 C. R. Seely, <https://orcid.org/0000-0001-5201-1236>
 J. A. A. McArt, <https://orcid.org/0000-0001-5654-9172>
 T. R. Overton <https://orcid.org/0000-0003-4098-4916>